Simulation-Based Optimization of Catastrophe Rescue with Quadruped Robots in Maze-Like Pathfinding Using Data Analytics

Simon Völkl, Sebastian Kleber, Tatyana Ivanovska
Fakultät Elektrotechnik, Medien und Informatik, Ostbayerische Technische Hochschule Amberg-Weiden,
Kaiser-Wilhelm-Ring 23, 92224 Amberg, Germany
s.voelkl2, s.kleber2, t.ivanovska@oth-aw.de

Keywords: Simulation, Optimization, Maze, Grid, Catastrophe Rescue, Data analytics, Graph, Pathfinding, Quadruped robot, robotic dog, Boston Dynamics Spot, Mesa, batch run, A Star, Manhattan Distance, Python, Symbolic AI

Abstract: This paper explores the potential of robotic systems like quadruped robots for catastrophe rescue operations through a simulation of grid-based maze environments. A bespoke framework was implemented, facilitating comprehensive data analysis on extensive batch runs. This analysis enabled the assessment of the impact of features on rescue efficiency. The A-Star algorithm with Manhattan Distance heuristics was implemented for pathfinding, while significant correlations between maze dimensions, rescue zone placement, agent coordination, and survivors were identified. The findings indicate that the allocation of sufficient rescue zones is imperative to ensure the minimization of transport pathways. Furthermore, an excess of agents can impede pathfinding performance if adequate coordination is lacking. The study indicates that the strategic distribution of rescue zones can substantially enhance rescue performance, attributable to the unique characteristics of isolated rescue zones. The study also evaluates the feasibility of deploying Boston Dynamics Spot quadruped robots and highlights operational challenges. These findings offer actionable insights for real-world disaster scenarios, emphasizing the optimization of rescue zone allocation and multi-agent coordination to enhance rescue operations.

1 INTRODUCTION

The efficient rescue of survivors is a key challenge in disaster scenarios. In this context, robotic rescue systems are becoming increasingly important. This is due to the fact that robots are easier to expend than humans in dangerous rescue operations, can also work in hazardous environments, and communicate more efficiently with each other. The objective of this thesis is to methodically examine the possible applications of such systems through simulation-based analyses in labyrinthine environments. A series of labyrinths were generated algorithmically, and their structural properties were evaluated. This analysis was used to draw conclusions regarding the optimal use of a rescue robot.

1.1 Related Work

Rahiman et al. (Gul et al., 2019) have previously examined a range of navigation techniques for both static and dynamic environments, with the potential for implementation in real-time navigation of mobile robots. It has been posited that the search algorithm A star is a suitable global navigation method for pathfinding.

Koval et al. (Koval et al., 2022) presented an evaluation of autonomous navigation using a Boston Dynamics Spot robot and multiple map and localization methods as part of ongoing research on navigation in GPS-denied environments, such as rescue missions.

1.2 Outline

The rest of the article is structured as follows. In Section 2, we introduce the overall implementation of the simulation, while Section 3 describes the data analysis, results and interpretations of the simulation batch runs. A possible real-world implementation with robot agents is discussed in Section 4, while also highlighting possible restrictions. Finally, Section 5 summarizes the article and gives an outlook on future work.

2 SIMULATION IMPLEMENTATION

The iterative approach of random DFS was selected as the algorithm for generating the maze, as the structure of the generated maze was conducive to the simulation. For instance, it corresponds to debris in disaster rescue operations. The exits of the maze are placed randomly at the edges. These zones function as secure environments where survivors can receive necessary care and support. The survivors are then randomly positioned within the maze, with the caveat that they are not to be placed in the positions of the safety zones. The robot search dog is placed randomly in one of the safety zones, from which it starts and can perform one of the following actions at each simulation step:

- move to the nearest survivor if no other robot moved there
- pick up survivor
- move to the nearest save zone
- drop off survivor

The action *pick up survivor* is indicative of the agent's successful identification of a survivor, thereby ensuring that the agent is not already carrying one. Conversely, the term *drop off survivor* signifies that the agent has a survivor and is in a safe zone. In the process of retrieving and lowering survivors, only those who have not yet been rescued are considered. In the context of movement, the A-Star algorithm is employed for pathfinding, with the NetworkX library serving as the underlying framework. This algorithm takes into account the previous path costs $g(n)$ and the heuristic function $h(n)$. The Manhattan distance is utilized as the heuristic. Due to the fact that movement in the maze or grid is constrained to horizontal and vertical steps, the Manhattan distance provides a more realistic and frequently more accurate estimation of the actual path length (for non-diagonal movements), which is an important criterion for the A-star algorithm. In each simulation step, the robots act sequentially and execute their action.

The objective of the simulation is to facilitate the movement of all survivors to the designated safety zones in a minimum number of steps and moved fields by the quadruped robot, thereby ensuring their access to necessary care. The implementation of the algorithm and batch run was executed in Python, leveraging the Mesa library. In addition, modules such as Pandas, Numpy, and Seaborn were utilized for the evaluation process. The simulation was executed a total of 58,800 times within a duration of one hour and 15 minutes, with a range of parameters varied as shown in Table 1.

The full source code is available at https://github.com/username/repo.

3 SIMULATION RESULTS AND DATA ANALYSIS

3.1 Correlation Heatmap

The correlation heat map, when analyzed with the Pearson coefficient, reveals discernible patterns, as shown below in Figure 1.

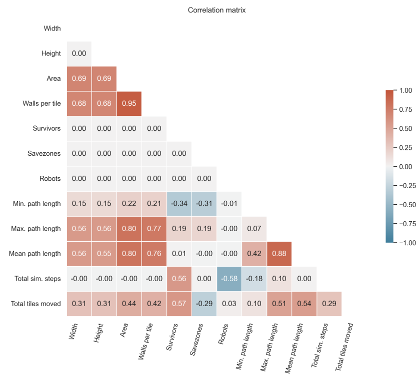

Figure 1: Correlated Heatmap.

It is evident that the number of total simulation steps is independent of the height, width, or quantity of save zones. The number of survivors to be rescued and the robots available are the only factors that influence the simulation steps. In the event that the range of movement of the robots per simulation step were to be limited, it would be possible to display one of the parameters in this location.

3.2 Influences on Path Lengths

The iterative approach of random DFS results in the generation of dead ends in larger areas of the maze with greater frequency. The optimal mazes, characterized by two exits (one at the start and one at the finish) and an absence of dead ends, exhibit a maximum path length between two points that is equivalent to the total number of squares in the maze. This is due to the fact that there is only one path from the entrance (point 1) to the exit (point 2).

It has been demonstrated that dead ends generally result in a more complex maze, yet they reduce the effective path length in comparison to a maze with a single passage. This phenomenon occurs because certain spaces that could be incorporated into the single, extensive passageway between the entrance and exit

Table 1: Simulation parameters and their respective values.

Parameter	Description	Values
Width	Width of the maze	8, 10, . . . , 20
Height	Height of the maze	8, 10, . . . , 20
Survivors	Number of survivors	1, 2, . . . , 10
Savezones	Number of rescue zones	1, 2, . . . , 10
Robots	Number of search robots	1, 2, . . . , 6

are instead utilized as dead ends, which are mostly not relevant for the pathfinding.

In the following analysis, the dead ends in the path between two points will be disregarded, as their effect on the path length is negligible. Consequently, the total maximum and average path length required will decrease. However, it should be noted that this decrease will not occur in a linear fashion, as will be demonstrated in the appendix 5.

Additionally, it has been observed that the minimum path lengths between save zones and survivors decrease as the number of save zones increases. This phenomenon can be attributed to the increased number of possibilities for shorter and thereby more optimal paths, as illustrated in the appendix 6. At the same time, the maximum path lengths increase due to a higher likelihood of encountering the longest and least optimal paths. However,the robot's operational efficacy is enhanced by the presence of a greater number of save zones in the context of disaster rescue missions. This augmentation is attributed to the increased availability of shorter, optimal paths that can be used.

3.3 Analysis of the Total Distance Moved

As indicated by the correlation matrix, the total distance traversed by the quadruped robots evidently increases with the height and width of the maze, as the distance from the safe zone to the survivor generally becomes greater. While the influence of the distance reduction due to the dead ends on the distance is discernible, it is generally negligible when compared to the effect of the increasing area.

The total distance traveled by the robots increases in proportion to the number of survivors, as demonstrated in the appendix 7. As the number of survivors increases, the minimal decrease in the graph's slope becomes slightly perceptible. This reduction in the number of survivors is due to the decreasing minimum path lengths between the robot search dogs, which originate from the safe zones, and the survivors. As the number of possible paths increases, so do the direct shortest paths for the search (see appendix 8).

It has been demonstrated that an increase in the number of safe zones is accompanied by a corresponding reduction in the distance traversed by robot search dogs, as it can be seen in the Figure 2. The implementation of additional safe zones has been demonstrated to result in a substantial decrease in the distance traversed. This reduction can offer a notable advantage in real-world scenarios.

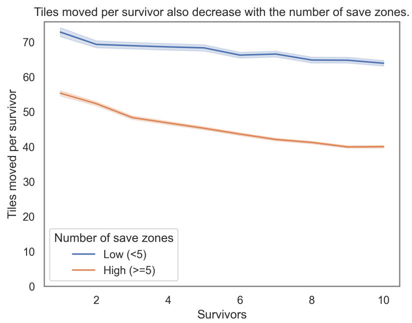

Figure 2: Total tiles moved per survivor per survivor count grouped by save zone count.

Conversely, an increase in the number of robot search dogs results in a slight decrease in the distance traveled (see appendix 9). This phenomenon can be attributed to the interference between search robots, which results in the disruption of each other's optimal paths to survivors. In a real-world implementation, it is therefore essential to prioritize effective communication and coordination between the robots to minimize search time. Consequently, the presence of robots has been observed to exert a negative influence on path lengths, resulting in a substantial augmentation of the original distances traversed.

The following recommendations for action in real-world disaster rescue can be derived from the above analyses: To mitigate these issues, it is imperative to establish a sufficient number of rescue zones at first. In order to prevent mutual obstruction during the process of rescuing survivors, it is essential to ensure that there is adequate communication and coordination within the rescue team. Furthermore, it is essential that different groups operate without interfering with one another's rescue efforts.

3.4 Process-Step-Analysis

Naturally, the number of simulation steps decreases as the number of robots increases, since multiple robots can rescue survivors simultaneously. In this context, the number of simulation steps is approximately halved when the number of robots is doubled—provided that the number of survivors is sufficiently large. This effect is clearly visible in Figure 3, particularly in the transition from one to two robots.

When the number of robots is very high — or even exceeds the number of survivors — the number of simulation steps approaches a value of four. This results from the sequence of the robots' minimum four required actions. In the extreme case, a single survivor is rescued by exactly one robot, which requires precisely four steps to complete the rescue. In real-world rescue scenarios, process steps should therefore be optimized as much as possible to accelerate the rescue of survivors. The number of rescue personnel is also a critical factor.

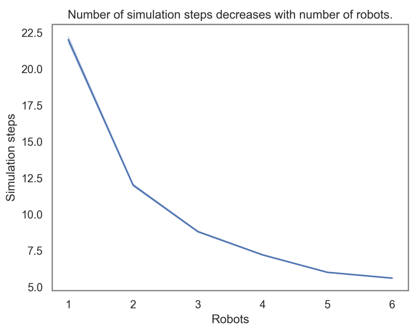

Figure 3: Total simulation steps per robot count.

3.5 Optimization of Save Zone Placement

A thorough examination of the simulations revealed that isolated save zones frequently contain a substantial number of survivors. Isolated save zones are defined as those that have a high or the highest path length to all other save zones. According to this definition, there must always be one save zone that is most isolated from all other save zones in the same maze. The analyses thus compared the minimum path length of each save zone to the other save zones in each simulation, including the number of survivors ultimately stationed in the zone.

The calculation of the numerous path lengths necessitated a substantial computational effort, yielding a single data point for each save zone. This data contained information regarding the maze, distance estimates and path lengths to other save zones, the number of survivors in the field, and general simulation data. A correlation heat map for the Pearson correlations can be found in the appendix 10.

Table 2 offers a concise representation of the most relevant values from the correlation matrix. It presents the linear correlations between the number of survivors stationed on the safe zone and each respective metric using Pearson's r, as well as the rank-based correlations using Kendall's τ and Spearman's ρ.

Given the linear nature of the relationship, the Pearson coefficient should exhibit superior accuracy in comparison to the Spearman's rank-based coefficient rho and Kendall's tau(Hauke and Kossowski, 2011). However, if the rank-based coefficients were to be utilized for comparison, no significant difference should be observed between them(Puth et al., 2015).

The correlation between the number of survivors stationed in the save zone and the minimum path length to other save zones is clearly evident (see appendix 11). The hypothesis that up to one additional survivor is located in isolated save zones is confirmed by the data. It is noteworthy that the median minimum path length exhibited a high deviation of 6.00, the average registered 12.76, and the standard deviation was particularly elevated at 18.55, as illustrated in the Table 3.

If a safe zone is placed at the greatest possible distance from other safe zones, it tends to accommodate more survivors on average by the end of the simulation than those that are not placed in isolation. More rescued survivors in a single safe zone indicate that this zone was more frequently selected as part of the most efficient path. Consequently, the total distance moved by the agents in the simulation is reduced, as more efficient paths can be selected. The optimization of save zones in the absence of information regarding the survivors' positions would thus be conducted as follows: It is imperative that the save zones be positioned in a manner that maximizes spatial separation between them to guarantee the most efficient paths to any survivor location.

The objective is to ascertain the combination of safe zone positions that optimizes the minimum path lengths to other safe zones. In the worst-case scenario, the simulation would place two safe zones in directly adjacent, accessible fields, which would essentially allow the two safe zones to be grouped into a single safe zone with a minimal deviation in distance, if any. In general, the following insights can be derived for a real-world scenario in labyrinth-like areas: In order to minimize the time and distance required in the rescue scenario, it is imperative that the rescue

Table 2: Correlation coefficients for distance-based metrics.

Metric	Pearson r	Kendall tau	Spearman rho
Euclidean Distance	0.19	0.17	0.21
Manhattan Distance	0.22	0.19	0.23
Minimum Path Length	0.21	0.20	0.25

Table 3: Summary statistics for distance-based metrics.

Metric	Median	Mean	Standard Deviation
Euclidean Distance	3.00	3.90	3.34
Manhattan Distance	3.00	4.38	4.10
Minimum Path Length	6.00	12.76	18.55

zones have paths that are as far apart as possible so that the route to the survivors can be optimized to the greatest extent possible.

In real-world scenarios, the structure of the disaster area—modeled here as a labyrinth—may not be known in advance, and only the size of the area might be available. Therefore, the placement of safe zones must be possible without initial pathfinding between the zones to be allocated. The following section demonstrates, based on a solid data foundation, that placing rescue zones with the greatest possible path lengths between them can be approximated using measured absolute distances as a rough visual estimate. For this purpose, both the grid-based Manhattan distance and the more realistic Euclidean distance are presented in Figure 4.

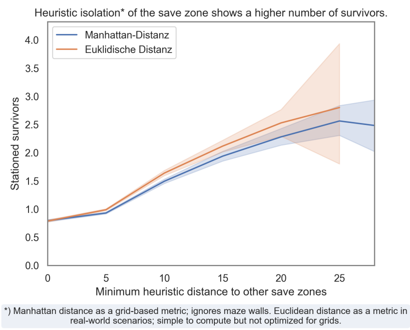

Figure 4: Total stationed survivor count per minimum heuristic distance to other save zones.

From the graphs showing the minimum Manhattan and Euclidean distances between the safe zones, it becomes evident that even a rough visual estimation of the distances between safe zones can be sufficient to maximize the minimum path lengths to other zones, thereby reducing the number of fields traveled by the agents. The grid-based Manhattan distance is particularly relevant for the simulation, as diagonal movement is not permitted. In contrast, real-world scenarios allow for more flexible movement directions, making the Euclidean distance more appropriate. Contrary to initial expectations during research, the Euclidean and Manhattan distances exhibit virtually identical correlation values, indicating no significant difference between the two. In cases where no information about the structure of the disaster area is available, safe zones should therefore be allocated as far apart from each other as possible, based on rough visual estimation.

The following conclusions can be drawn for a real-world scenario based on the findings from the simulations and the evaluation of the safe zones:

If the structure of the disaster area is known — such as in the simulated labyrinth — safe zones should be positioned so that the paths between them are as long as possible. This allows agents to find more efficient routes and reduces search time.

If the structure of the area is not known in advance, safe zones can still be placed based on rough visual estimation, using the largest possible distances between them. For grid-based environments, distances should be calculated using the Manhattan distance; otherwise, the Euclidean distance is more appropriate.

Under no circumstances should safe zones be placed in clusters or close proximity to one another, as they could be interpreted as a single zone from a pathfinding perspective. Instead, resources should be used to position safe zones in areas not yet covered. This approach effectively reduces rescue time and transport distances.

4 REAL-WORLD IMPLEMENTATION

In real-world scenarios, disaster rescue operations are significantly more complex and difficult than in grid-based simulations without hindering external factors. To facilitate the effective rescue of survivors, it is imperative to employ state-of-the-art robot dogs that are meticulously engineered for rescue operations. The optimal quadruped robot would require the following equipment and characteristics:

- Precise sensor technology with edge processing for the detection and recognition of obstacles, hazards, shortcuts, and survivors.
- Integrated dynamic map for reliable, adaptable, and short pathfinding.
- Built-in mechanisms for pulling or carrying survivors, such as a robotic arm.
- Coordination and communication tools for deployment in a fleet.

The successful robotic dog Spot by Boston Dynamics demonstrates impressive reliability in several areas. In addition to its sophisticated sensor technology, it also supports map integration, which enables the robot to avoid obstacles and identify optimal paths when used in combination(Dynamics, 2025c; Koval et al., 2022).

As an additional feature for pulling loads, the Spot Arm can be used(Dynamics, 2020). According to the specifications, the robotic arm can pull a maximum weight of up to 25 kg on a carpeted surface. Based on this weight limit, it would be possible to pull a child weighing up to 25 kg — approximately the average weight of a seven-year-old — if placed on a stretcher and pulled by the robotic dog's arm(Stolzenberg et al., 2007). Promotional videos from Boston Dynamics also show ten Spot robots simultaneously pulling a truck from a standstill at a slow pace(arronlee33, 2019). This suggests that transporting heavier survivors may also be feasible, especially if the stretcher is equipped with wheels.

These transport capabilities — assuming similar specifications — can also be applied to other robotic dogs or robots. However, the movement of robots under significant pulling loads on wheeled stretchers still requires validation. Nonetheless, the use of robotic search dogs for locating and transporting survivors in disaster scenarios appears to be a viable option.

For coordination and communication between robotic dogs, various software solutions from the manufacturer can be used in combination. Orbit is a fleet management and data analysis software designed for operating multiple Boston Dynamics robots(Dynamics, 2025a). Additionally, the Spot SDK enables the development of custom applications(Dynamics, 2025b). Together, these tools allow for synchronized coordination and communication, with integrated data analysis and management. Other solutions—whether manufacturer-specific or independent—also support software-based orchestration and control of robots.

5 SUMMARY AND FUTURE WORK

This paper explores the potential of robotic systems, particularly Boston Dynamics' Spot, for efficient rescue operations in disaster scenarios using grid-based maze simulations. A custom simulation environment was developed through the utilization of Python and Mesa to emulate labyrinthine terrains, thereby simulating real-world disaster zones. The A-Star pathfinding algorithm with Manhattan distance heuristics was employed to optimize robot navigation.

Key parameters such as maze dimensions, number of survivors, rescue zones, and robot agents were varied across batch simulations. The analysis revealed correlations between structural features and rescue efficiency, underscoring the significance of the correct rescue zone placement and agent coordination. It is noteworthy that isolated rescue zones consistently hosted a greater number of survivors, while a wider spatial distribution of zones contributed to reduced travel distances and fewer simulation steps. The study focussed on actionable insights for real-world implementation, emphasizing strategic save zone allocation and robot fleet coordination to enhance rescue effectiveness.

Future research should aim to validate the simulation results through physical testing with real robotic agents, such as Boston Dynamics' Spot. These experiments should assess the robot's operational performance under realistic conditions, with additional attention to energy consumption and battery life — both of which are critical parameters in prolonged rescue missions.

In addition, the deployment and evaluation of multi-agent communication software should be conducted in real-world environments to enhance coordination and minimize interference among multiple rescue units.

Practical trials may also involve equipping Spot with a robotic arm and a transport platform, such as a wheeled stretcher, to facilitate the evacuation of survivors. These configurations should be tested using anthropomorphic dummies of varying weight classes to determine the feasibility of transporting human-like payloads. Furthermore, such tests should be conducted in challenging terrain to simulate realistic disaster conditions. This includes environments with steep inclines, scattered debris, and uneven surfaces, which reflect the operational complexities typically encountered in real-world rescue scenarios. These evaluations will help identify mechanical limitations and inform operational strategies for deploying robotic agents in disaster response missions.

TRANSPARENCY INFORMATION

In accordance with the European Code of Conduct for Research Integrity: Language models (GPT-4.1, GPT-5) were utilized as instruments in the production, formulation, and revision of the work. The translation processes also incorporated the use of DeepL.

ACKNOWLEDGEMENTS

We gratefully acknowledge Prof. Dr. Phil. Tatyana Ivanovska for her invaluable guidance and support throughout this project. We also gratefully acknowledge the OTH Amberg-Weiden for providing the research funding that made this work possible.

REFERENCES

arronlee33 (2019). Boston dynamics - spotmini robots pulling truck tests [1080p]. https://www.youtube.com/watch?v=PBukEnjTJfc. YouTube video, last accessed 28 August 2025.

Dynamics, B. (2020). Spot arm. Last accessed 10. June 2025.

Dynamics, B. (2025a). Boston dynamics orbit. Last accessed 28. August 2025.

Dynamics, B. (2025b). Boston dynamics spot sdk. Last accessed 28. August 2025.

Dynamics, B. (2025c). Components of navigation. Last accessed 10. June 2025.

Gul, F., Rahiman, W., and Nazli Alhady, S. S. (2019). A comprehensive study for robot navigation techniques. *Cogent Engineering*, 6(1):1632046.

Hauke, J. and Kossowski, T. (2011). Comparison of values of pearson's and spearman's correlation coefficients on the same sets of data. *Quaestiones geographicae*, 30(2):87–93.

Koval, A., Karlsson, S., and Nikolakopoulos, G. (2022). Experimental evaluation of autonomous map-based spot navigation in confined environments. *Biomimetic Intelligence and Robotics*, 2(1):100035.

Puth, M.-T., Neuhäuser, M., and Ruxton, G. D. (2015). Effective use of spearman's and kendall's correlation coefficients for association between two measured traits. *Animal Behaviour*, 102:77–84.

Stolzenberg, H., Kahl, H., and Bergmann, K. (2007). Körpermaße bei kindern und jugendlichen in deutschland. *Bundesgesundheitsblatt - Gesundheitsforschung - Gesundheitsschutz*, 50:659–669.

APPENDIX

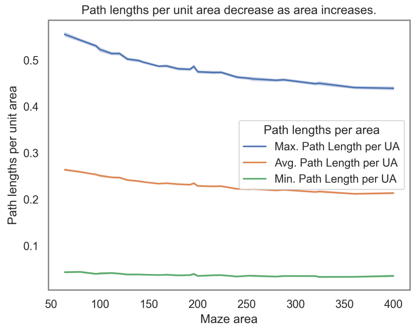

Figure 5: Path lengths per area.

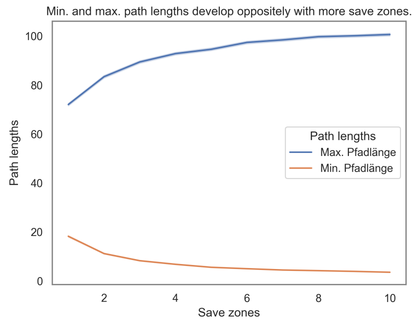

Figure 6: Path lengths per save zone count.

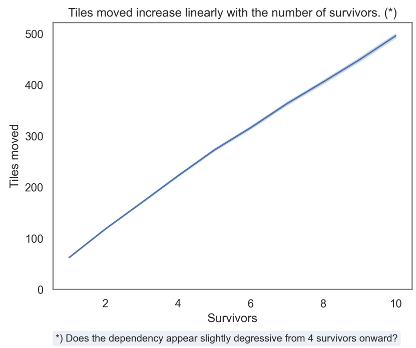

Figure 7: Total tiles moved per survivor count.

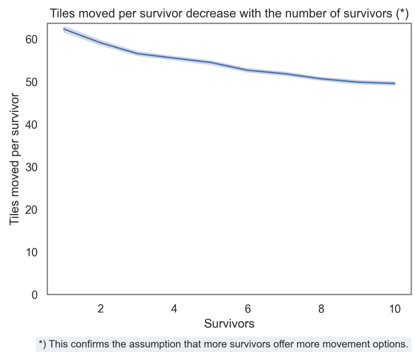

Figure 8: Total tiles moved per survivor per survivor count.

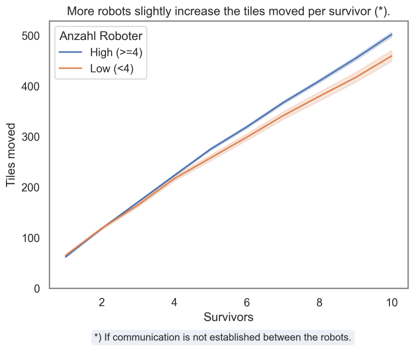

Figure 9: Total tiles moved per survivor per survivor count grouped by robot count.

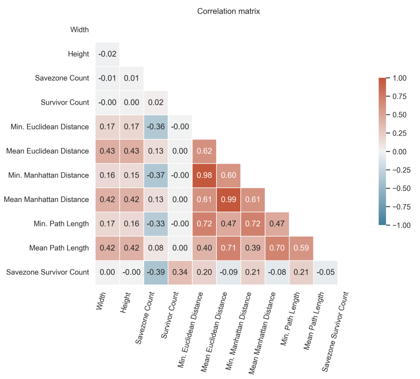

Figure 10: Correlation Heatmap for the save zone optimization.

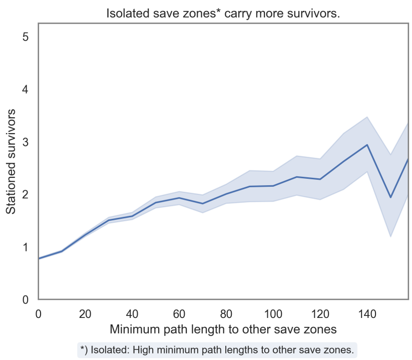

Figure 11: Stationed survivor count per save zone isolation.